

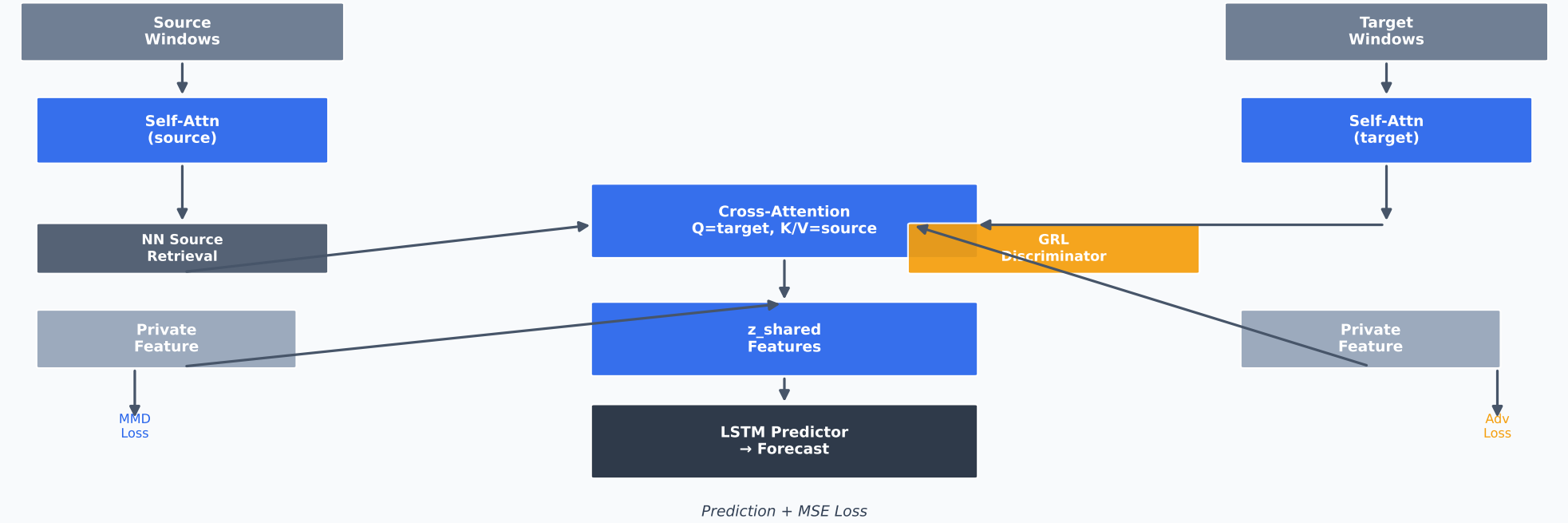
Cloud Workload Prediction: Method Comparison

CWPDDA · MC-CWPDDA · MCTL · Tr-Predictor

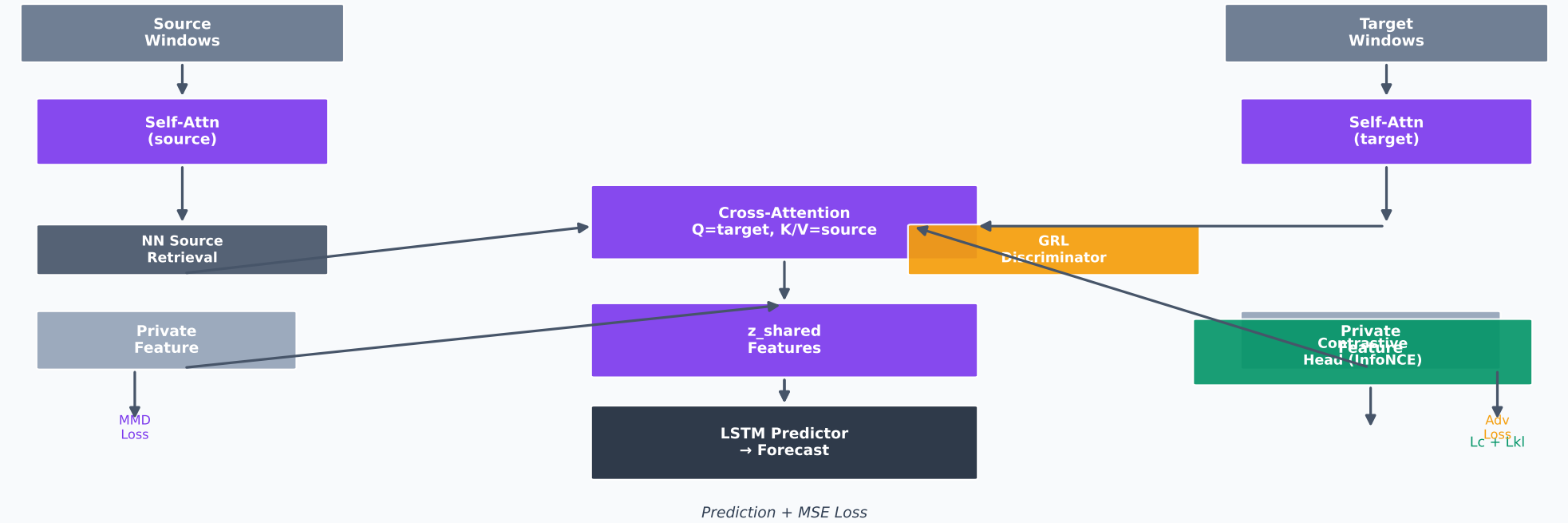
Attribute	CWPDDA	MC-CWPDDA	MCTL	Tr-Predictor
Problem	CPU/Memory prediction (Google→Alibaba)	CPU/Memory prediction (hybrid domains)	Few-shot workload (Google→Alibaba)	Small-sample workload (any→any)
Transfer Strategy	Adversarial + MMD disentanglement	Curriculum (pretraining → contrastive → joint)	Contrastive KL alignment (frozen source)	Two-stage TrAdaBoost reweighting
Supervision Level	Full supervision (source + target labels)	Full supervision (3-stage curriculum)	Few-shot (minimal target labels)	Full supervision (source heavy, target sparse)
Learning Paradigm	Zero-shot transfer at test time (nearest-neighbour source retrieval)	Curriculum learning (staged fine-tuning)	Few-shot contrastive learning	Instance-based transfer (sample reweighting)
Core Architecture	3-branch attention + LSTM predictor + GRL discriminator	3-branch attention + contrastive head + GRL discriminator	TCN encoder + regression head (source frozen)	Ensemble of LSTM weak learners
Loss Function	MSE + MMD disentanglement + adversarial (GRL)	MSE + MMD + GRL + InfoNCE contrastive + KL	InfoNCE contrastive + KL divergence	Weighted MSE (AdaBoost.R2 sample weights)
Key Innovation	Cross-attention K/V = NN source retrieval	Combines CWPDDA + MCTL via curriculum stages	Mixup + PAPN for distribution alignment	Stage 1: boost target; Stage 2: decay source
Source Selection	All source data (no ranking)	All source data (no ranking)	All source data (no ranking)	TWED + Transfer Entropy ranking (top-k selection)

Architecture Diagrams

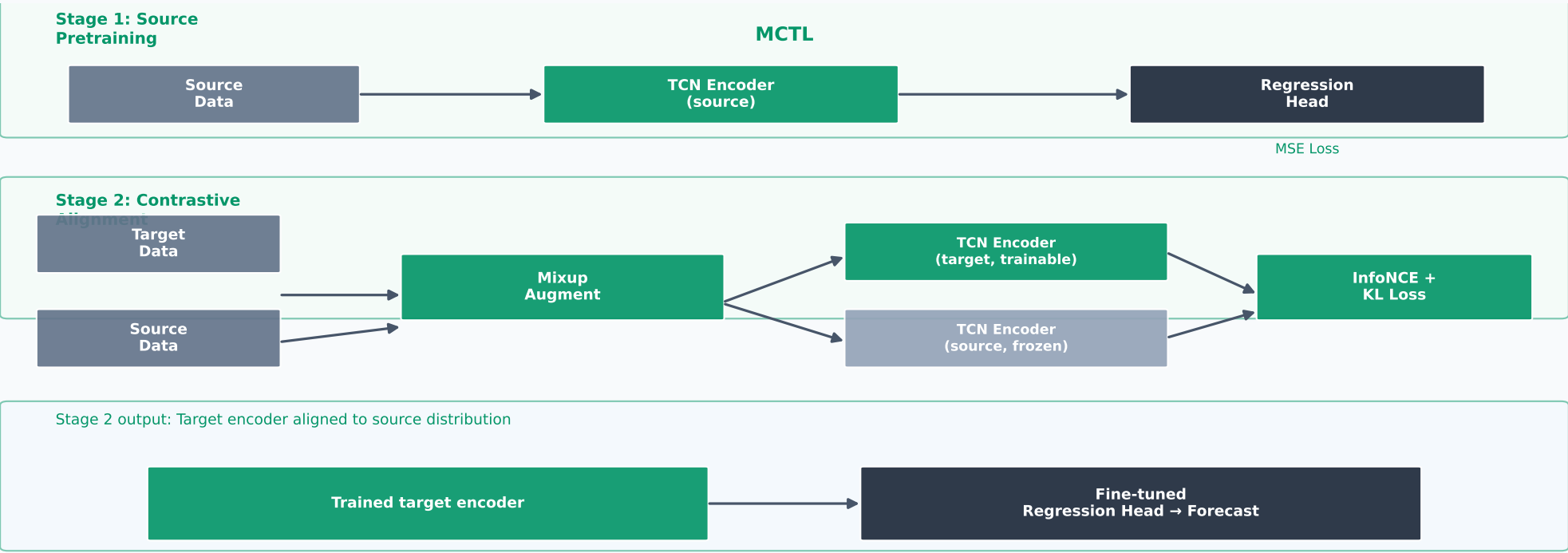
CWPDDA



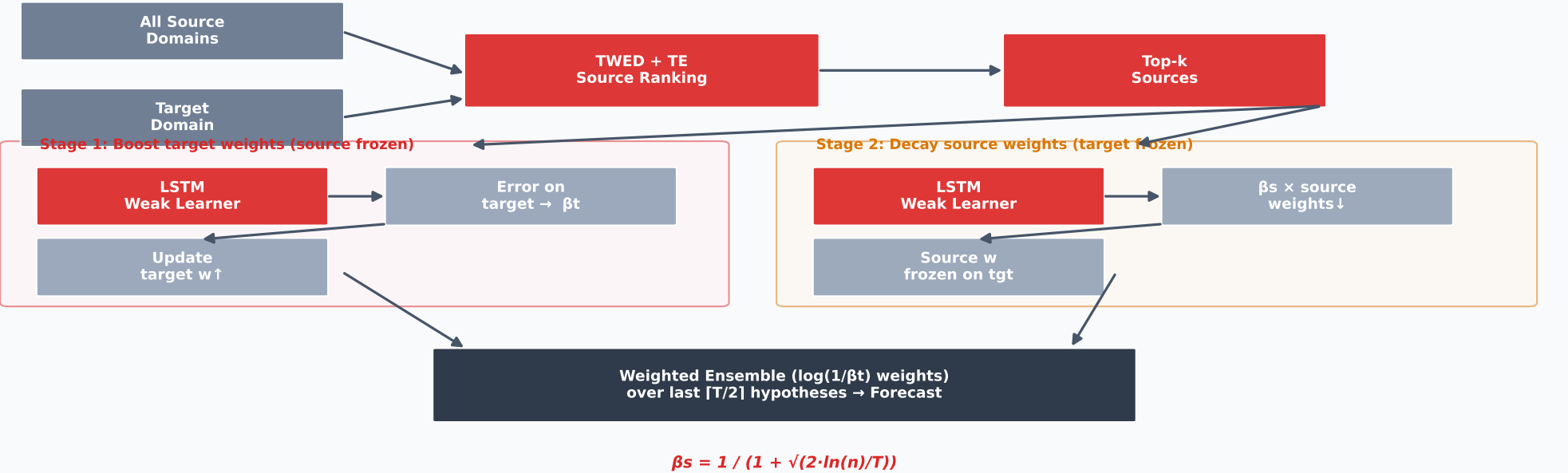
MC-CWPDDA



MCTL

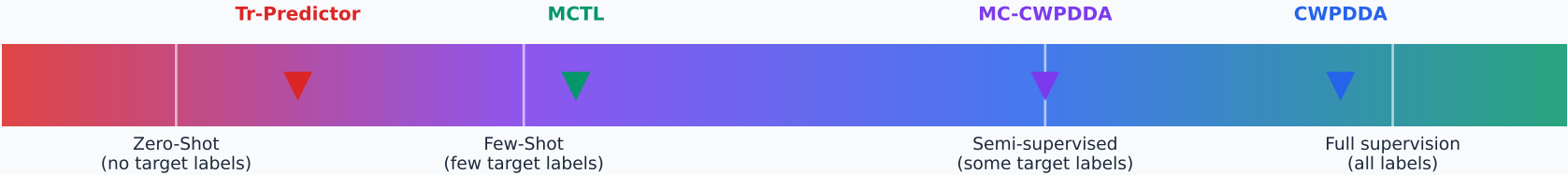


Tr-Predictor (TrAdaBoost.R2-LSTM)

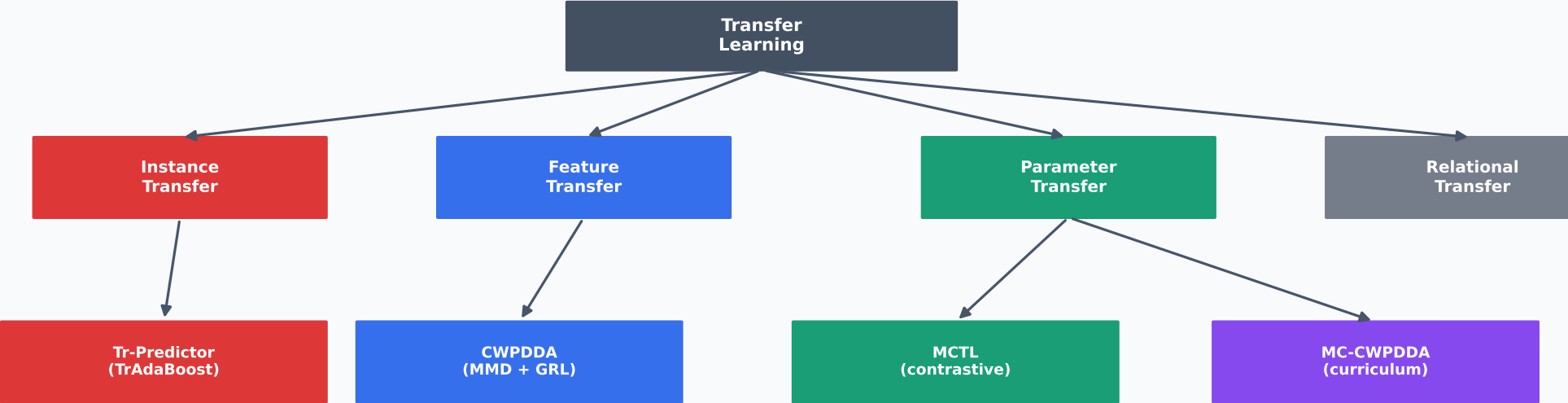


Transfer Learning Paradigms & Few-Shot Spectrum

Supervision Spectrum: Source Labels vs Target Labels



Transfer Learning Taxonomy

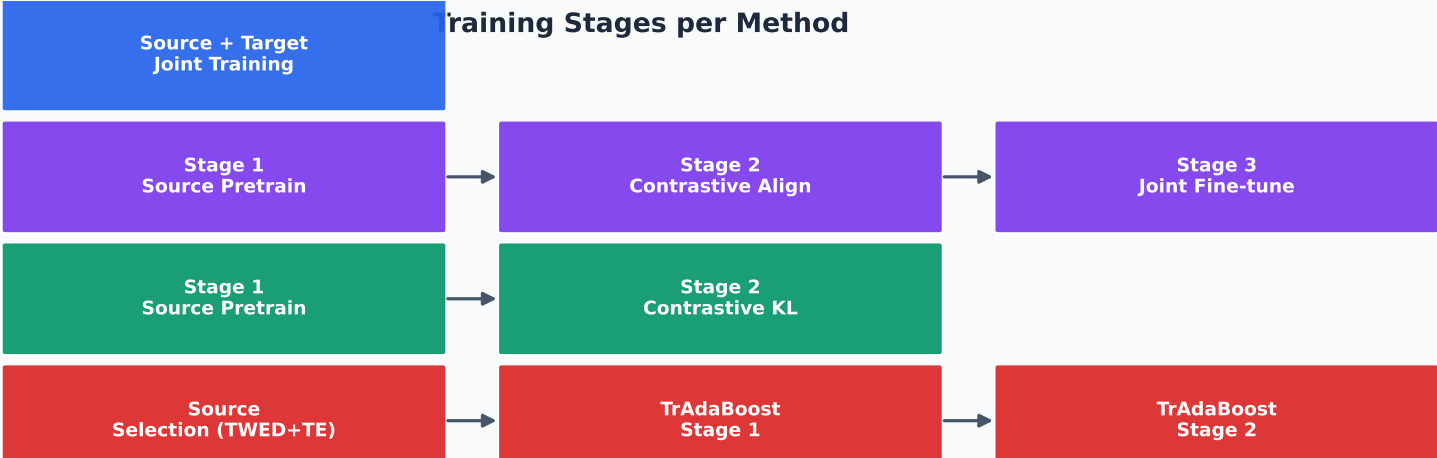


CWPDDA

MC-CWPDDA

MCTL

Tr-Predictor



Algorithm Pseudocode

Algorithm 1: CWPDDA — Domain-Adversarial Workload Prediction

Input: Source windows X_s , labels Y_s ; Target windows X_t
Output: Forecast $\hat{y}_t$ for target domain

- FOR each training step:
- Retrieve nearest-neighbour source window $\tilde{x}_s \leftarrow \text{NN}(x_t, X_s)$ [L2]
- $z_{\text{src_priv}} \leftarrow \text{SelfAttn}(\text{proj_src}(X_s))$
- $z_{\text{tgt_priv}} \leftarrow \text{SelfAttn}(\text{proj_tgt}(X_t))$
- $z_{\text{shared}} \leftarrow \text{CrossAttn}(Q=X_t, K=\tilde{x}_s, V=\tilde{x}_s)$
- $L_y \leftarrow \text{MSE}(\text{LSTM}(z_{\text{shared}}), Y_s)$ [prediction]
- $L_f \leftarrow -\text{mmd}(z_{\text{priv}}, z_{\text{shared}})$ [disentangle]
- $L_d \leftarrow \text{GRL_discriminator}(z_{\text{src_priv}}, z_{\text{tgt_priv}})$ [adversarial]
- $L \leftarrow L_y + \lambda_1 \cdot L_f + \lambda_2 \cdot L_d$
- Backpropagate L ; update all parameters
- RETURN $\text{LSTM}(\text{CrossAttn}(Q=X_{t_test}, K=\tilde{x}_s, V=\tilde{x}_s))$

Algorithm 2: MC-CWPDDA — Curriculum Contrastive Domain Adaptation

Input: Source (X_s, Y_s) , Target (X_t, Y_{t_few})
Output: Forecast $\hat{y}_t$

- ```
// Stage 1: Source encoder pretraining
1. Train {proj_src, SelfAttn_src} with MSE(LSTM(z_src), Y_s)

// Stage 2: Contrastive alignment (source frozen)
2. FREEZE source branch parameters
3. FOR each batch:
4. $\lambda \sim \text{Beta}(\alpha, \alpha)$
5. $x_{\text{mix}} \leftarrow \lambda \cdot x_s + (1-\lambda) \cdot x_t$ [cross-domain mixup]
6. $h_{\text{anc}} \leftarrow \text{normalize}(\text{ContrastiveHead}(\text{CrossAttn}(x_{\text{mix}})))$
7. $h_{\text{pos}} \leftarrow \lambda \cdot G_c(z_{\text{shared_src}}) + (1-\lambda) \cdot G_c(z_{\text{shared_tgt}})$
8. $L_c \leftarrow \text{InfoNCE}(h_{\text{anc}}, h_{\text{pos}}, \{h_{\text{neg_k}}\})$ [K=8 negatives]
9. $L_{kl} \leftarrow \text{KL}(q_{\text{tgt}} || q_{\text{src}})$ [distribution align]

// Stage 3: Joint fine-tuning
10. UNFREEZE all parameters
11. $L \leftarrow L_y + \lambda_1 \cdot L_f + \lambda_2 \cdot L_d + \lambda_3 \cdot L_c + \lambda_4 \cdot L_{kl}$
12. Train until early-stop on validation MSE
```

## Algorithm 3: MCTL — Mixed Contrastive Transfer Learning

Input: Source  $(X_s, Y_s)$ , Target  $X_t$  (few-shot, minimal labels)  
Output: Forecast  $\hat{y}_t$

- ```
// Stage 1: Train source TCN encoder
1. Train TCN_src end-to-end: min MSE(RegrHead(TCN_src(X_s)), Y_s)
2. FREEZE TCN_src

// Stage 2: Align target encoder to source distribution
3. FOR each batch (x_s, x_t):
4.    $\lambda \sim \text{Beta}(\alpha, \alpha)$  [ $\alpha=1.0$ ]
5.    $x_{\text{mix}} \leftarrow \lambda \cdot x_s + (1-\lambda) \cdot x_t$  [mixup]
6.    $z_{\text{mix\_s}} \leftarrow \text{TCN\_src}(x_{\text{mix}})$  [frozen]
7.    $z_{\text{mix\_t}} \leftarrow \text{TCN\_tgt}(x_{\text{mix}})$  [trainable]
8.   Sample K=32 in-batch negatives
9.    $p_s \leftarrow \text{PAPN}(z_{\text{mix\_s}}, z_{\text{pos}}, \{z_{\text{neg}}\}) / \tau$  [InfoNCE,  $\tau=0.1$ ]
10.   $p_t \leftarrow \text{PAPN}(z_{\text{mix\_t}}, z_{\text{pos}}, \{z_{\text{neg}}\}) / \tau$ 
11.   $L_{kl} \leftarrow p_s \cdot \log(p_s/p_t) + (1-p_s) \cdot \log((1-p_s)/(1-p_t))$ 
12.  Backprop  $L_{kl}$ ; update TCN_tgt only

// Inference
13.  $\hat{y}_t \leftarrow \text{RegrHead}(\text{TCN\_tgt}(X_t))$ 
```

Algorithm 4: Two-Stage TrAdaBoost.R2-LSTM (Tr-Predictor)

Input: Source data T_s (n samples), Target data T_t (m samples), T rounds
Output: Ensemble forecast $\hat{y}_t$

- ```
// Pre-step: Source domain selection
0. FOR each candidate source domain s :
 score(s) $\leftarrow \text{rank}(\text{TWED}(\text{tgt}, s)) + \text{rank}(-\text{TE}(s-\text{tgt}))$
 SELECT top-k sources by score

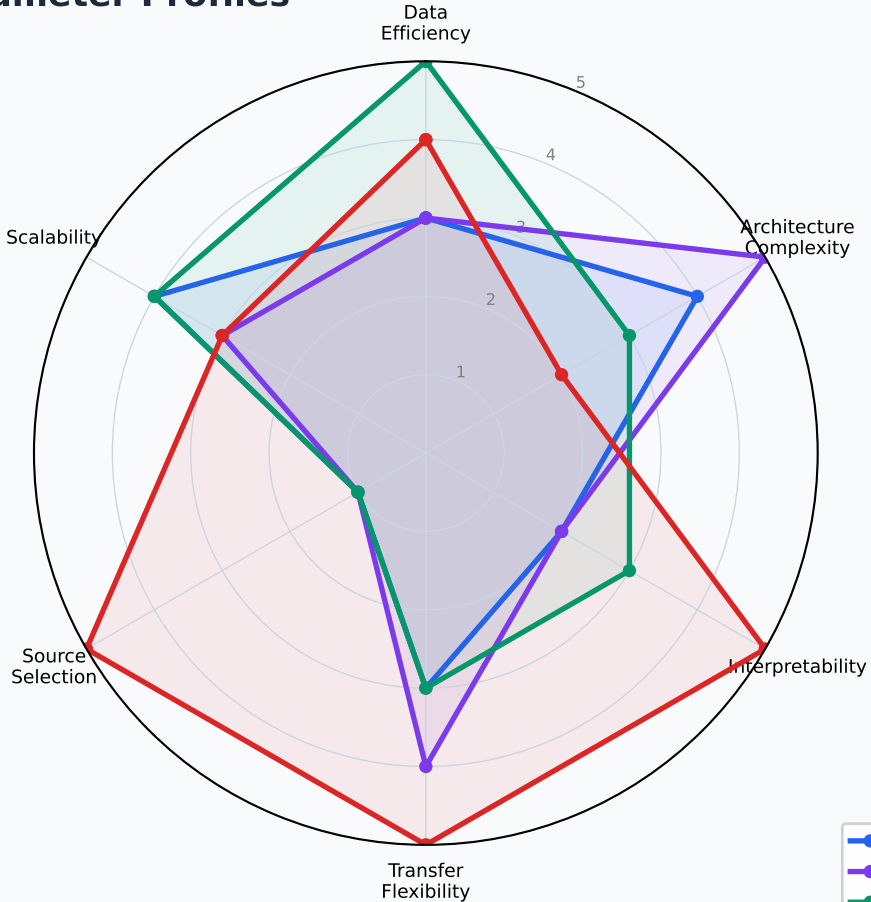
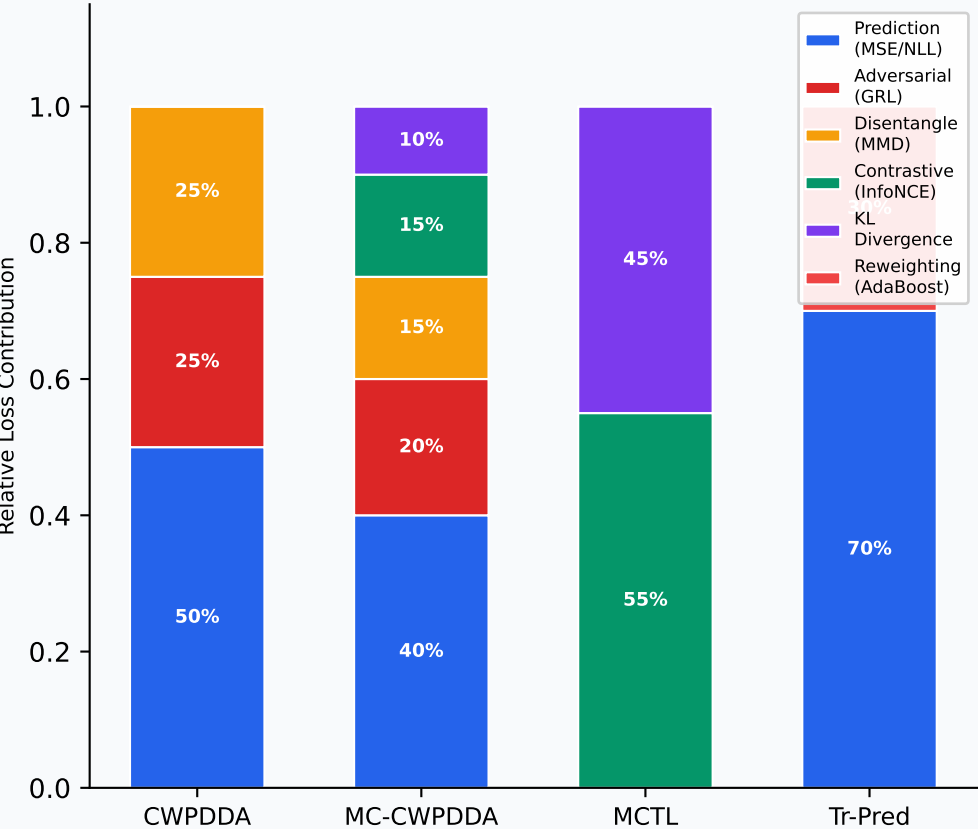
// Boosting
1. $w \leftarrow \text{uniform}(1/N)$ where $N = n + m$
2. $\beta_s \leftarrow 1 / (1 + \sqrt{(2 \cdot \ln(n))/T})$
3. FOR $t = 1, \dots, T$:
4. $h_t \leftarrow \text{LSTM trained on } (T_s \cup T_t) \text{ with weights } w$
5. $e_t \leftarrow \sum_{i \in T_t} w_i \cdot |h_t(x_i) - y_i| / \max|\text{error}|$
6. $\beta_t \leftarrow e_t / (1 - e_t)$
7. IF $t \leq \lceil T/2 \rceil$: [Stage 1 – target weight update]
8. $w_i \leftarrow w_i \cdot \beta_t^{(1 - |\text{error}_i|/D_{\text{max}})}$ $\forall i \in T_t$
 # source weights unchanged
9. ELSE: [Stage 2 – source weight decay]
10. $w_i \leftarrow w_i \cdot \beta_s$ $\forall i \in T_s$
 # target weights unchanged
11. Renormalise w

12. $\hat{y} \leftarrow \sum_{t=\lceil T/2 \rceil+1}^T \log(1/\beta_t) \cdot h_t(x)$ [weighted average]
```

Loss Functions & Hyperparameter Profiles

Method Profile (1=low, 5=high)

Loss Component Breakdown

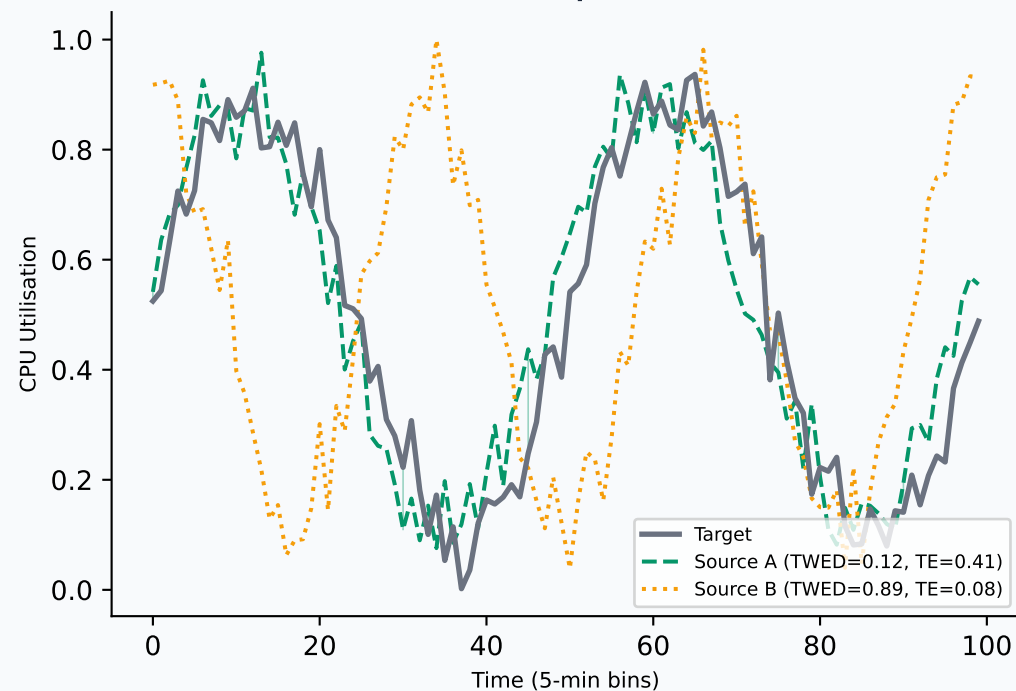


Key Hyperparameters

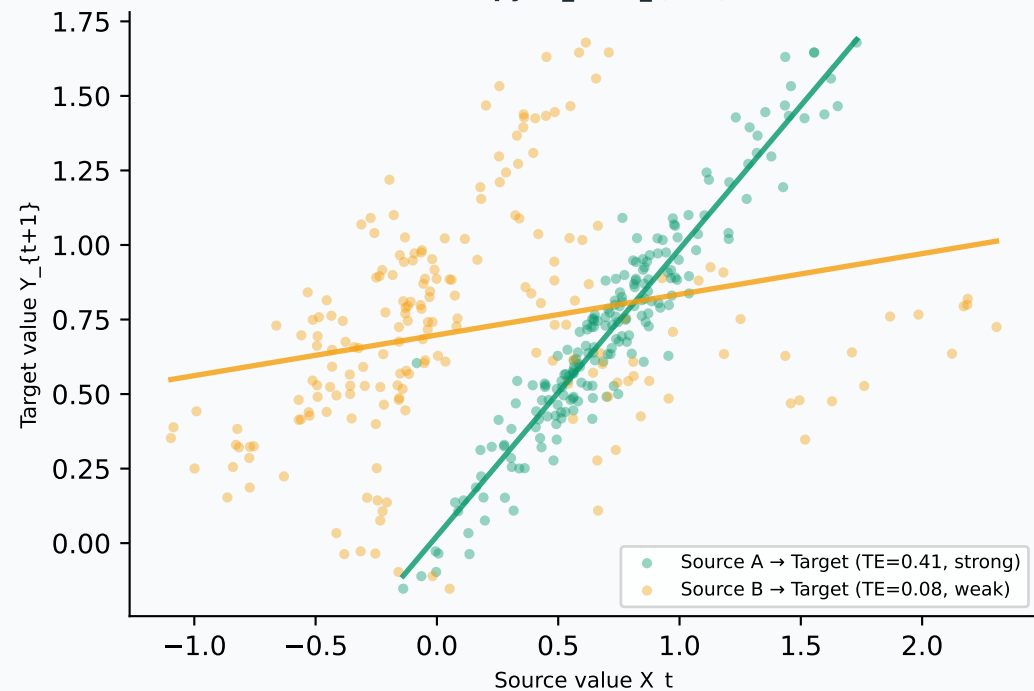
| Hyperparameter         | CWPDDA                     | MC-CWPDDA              | MCTL      | Tr-Predictor       |
|------------------------|----------------------------|------------------------|-----------|--------------------|
| Window size            | 24                         | 24                     | 24        | 24                 |
| Hidden dim / units     | 64                         | 64                     | 128 (TCN) | 64 (LSTM)          |
| Forecast horizon       | 1 step                     | 1 step                 | 1 step    | 1 step             |
| Dropout                | 0.1                        | 0.1                    | 0.2       | 0.1                |
| GRL $\alpha$ / $\beta$ | 10, 0.75                   | 10, 0.75               | —         | —                  |
| Loss weights $\lambda$ | $\lambda_1=\lambda_2=0.01$ | $\lambda_1-4=0.1/0.05$ | —         | $\beta_s = f(n,T)$ |
| Mixup $\alpha$         | —                          | 1.0                    | 1.0       | —                  |
| Contrastive $\tau$     | —                          | 1.0                    | 0.1       | —                  |
| Negatives K            | —                          | 8                      | 32        | —                  |
| Boosting rounds T      | —                          | —                      | —         | 20                 |
| Source selection       | None                       | None                   | None      | TWED+TE top-5      |
| Optimiser              | Adam 1e-3                  | Adam 1e-3              | Adam 1e-3 | Adam 1e-3          |

# Tr-Predictor Detail: Source Selection + Weight Dynamics

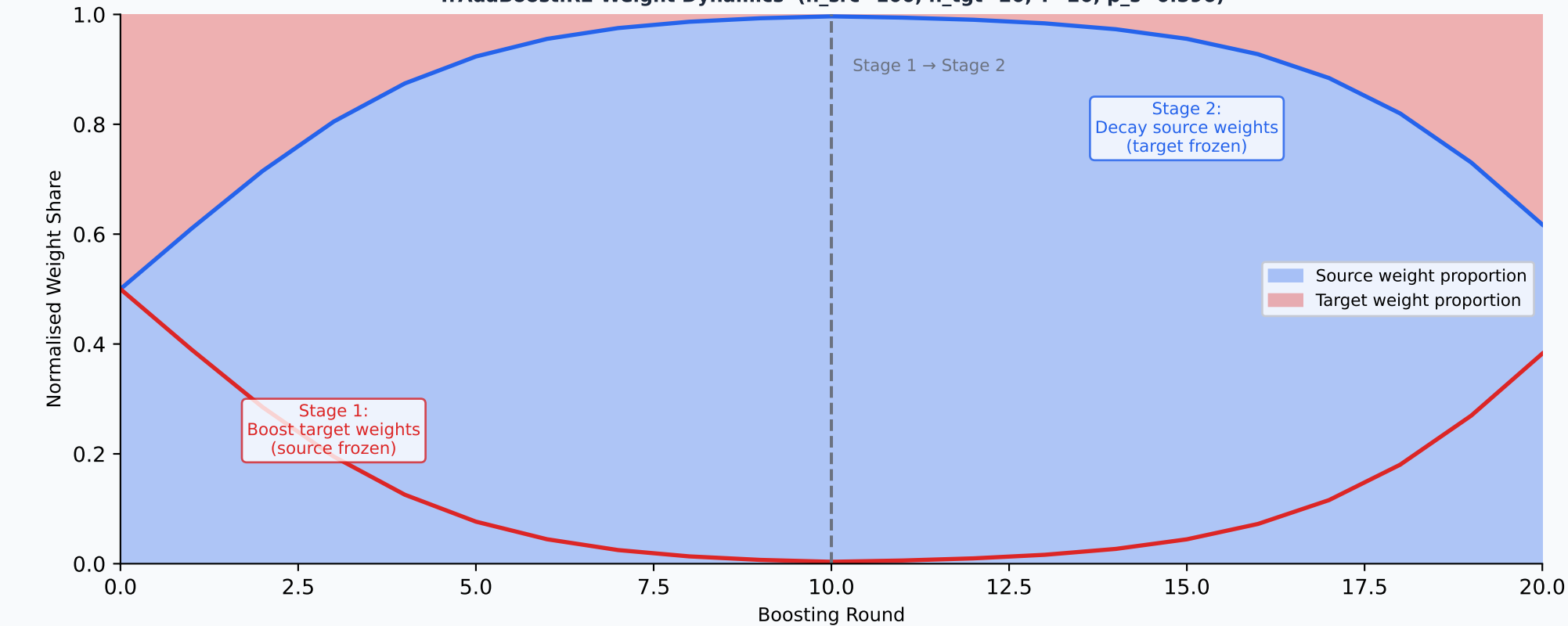
**TWED: Time Warp Edit Distance**



**Transfer Entropy:  $X_t \rightarrow Y_{t+1}$  influence**

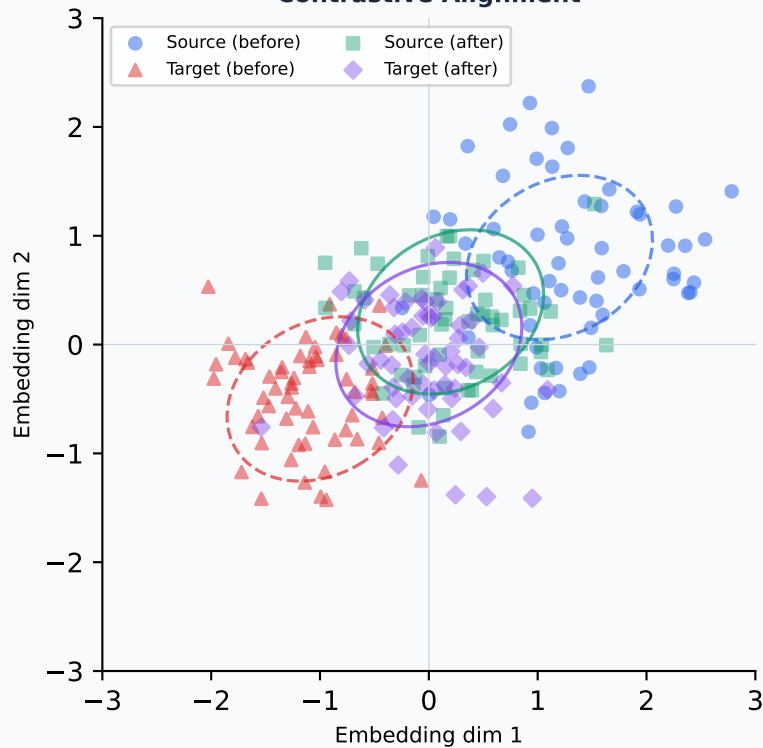


**TrAdaBoost.R2 Weight Dynamics ( $n_{src}=100$ ,  $n_{tgt}=20$ ,  $T=20$ ,  $\beta_s=0.596$ )**

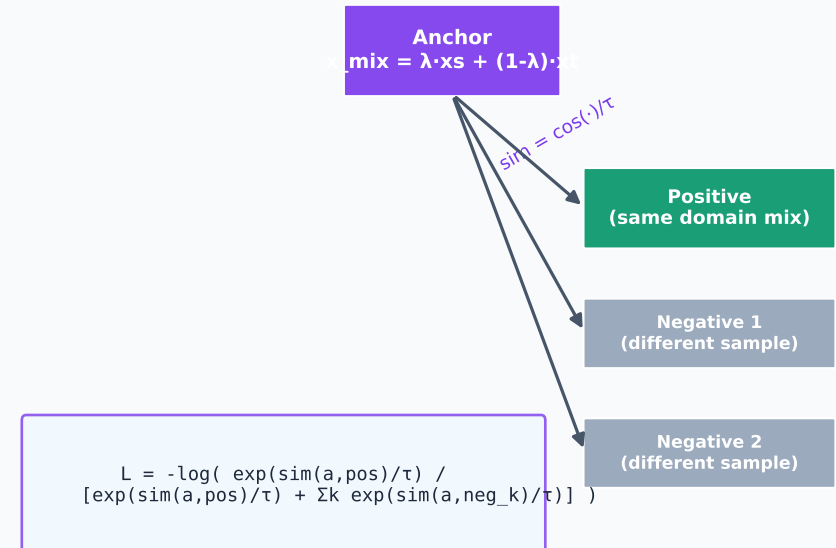


# MC-CWPDDA & MCTL: Contrastive Learning Detail

Feature Space: Before vs After Contrastive Alignment



InfoNCE / PAPN Contrastive Loss



MC-CWPDDA Curriculum Training Timeline

